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CSCU9M6 Natural Language Processing and Computer Vision (NLP&CV) Assignment Spring 2026

Due May 8, 2026

Sign Language Recognition Using Deep Convolutional Neural Networks

Total Marks: 100

Introduction:

Hand-gesture recognition using sign language is an important tool for enabling communication for the hearing-impaired and for building inclusive human–computer interaction systems. These systems rely on visual input (such as hand gestures) interpreted through machine learning models, particularly convolutional neural networks (CNNs), to identify signs representing letters or words. This assignment uses the Sign Language MNIST dataset, which contains grayscale images representing American Sign Language (ASL) alphabet signs. Each image corresponds to a hand gesture representing a letter (A–Y, excluding J and Z). The dataset provides a practical foundation for building and evaluating deep learning models capable of recognizing hand gestures for applications such as assistive communication tools and sign-language translation systems.

Project Objective: The objectives of this project are to develop and evaluate deep Convolutional Neural Network (CNN) models for classifying hand gestures in the Sign Language MNIST dataset.. The goals include:

1. Designing and implementing a baseline CNN architecture for sign recognition.
2. Enhancing model performance by incorporating advanced techniques and deeper, more modern convolutional network architectures (e.g., VGG, AlexNet, ResNet).
3. Analyzing the impact of architectural choices on model accuracy, robustness, and generalization.
4. Demonstrating the practical relevance of hand gesture recognition in inclusive technologies.

Dataset: The original dataset with more details is available here:

- **Name** Sign Language MNIST
- **Source:** https://www.kaggle.com/datasets/datamunge/sign-language-mnist
- **Classes**: 24 letters of the alphabet (A–Y, excluding J and Z)
- **Image Dimensions**: 28x28 pixels (grayscale)
- **Training Set**: 27,455 images
- **Test Set**: 7,172 images
- **Challenges**: Visual similarity between signs (e.g., D vs. E), low-resolution input, subtle intra-class variations.

Deliverables:

1. Data Loading and Preprocessing

Step	Description
1.1 Data Loading	Load CSV files and convert pixel values into image tensors
1.2 Preprocessing	Normalize pixels to [0, 1], and Create a validation split from the training set
1.4 ……….	…….
1…. Justification	Explain why resizing, normalisation, and splitting improve model training.

Outlining the preprocessing steps applied to the dataset, covering aspects such as data splitting (into training, testing, and validation sets), normalization techniques, reshaping of input data, and any other preprocessing methods utilized; etc. Justify the chosen preprocessing techniques and explain how they contribute to the overall quality and efficiency of the model training process.

2. CNN Architectures

2.1. Model	**Architecture**
2.2. Simple CNN	Define a baseline CNN (e.g., Conv2D, MaxPool, Conv2D, MaxPool, Dense layers).
2.3. Enhanced CNN	Enhanced CNN architectures (e.g., deeper CNN, VGG-style network).

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3. Training, testing, and evaluation, including hyperparameters tuning [30 marks]

Component	Description
3.1 Hyperparameters	E.g., batch size, epochs, cost-function, optimizer, learning rate, weight initialisation, activation function, regularization, and dropout.
3.2 Training Curves and tables	E.g.Plot training vs. validation accuracy/loss using matplotlib. Compare accuracy, loss, and training time for models. Compare accuracy, loss, and training time for models.
3.3 Confusion Matrix	Generate and visualise e.g., using Seaborn library.
3.4 ……	……..
3.	Classification report and predicted label, include precision, recall, and F1-score. Sample predictions. Show several test images with the true label

Describe the process of training, testing, and evaluating the CNN models, including details on hyperparameter tuning techniques utilized to optimize model performance. Present all obtained results (including those related to hyperparameter tuning) using a summary table. Explain how hyperparameters such as learning rate, batch size, optimizer choice, and regularization methods were adjusted and justified during the training phase. Provide insights into the evaluation metrics used to assess the models' performance on the testing and validation datasets.

4. Reflection

Discuss model limitations, dataset limitations, overfitting/underfitting, possible improvements, cite one peer-reviewed reference paper.

You are required to submit both the code and a report that explains the four deliverables detailed above.

Submission Guidelines

- Code: Jupyter Notebook with complete implementation.
- Report: (1500–2000 words) PDF document covering all deliverables.
- Submission deadline: May 8, 2026.

Examples of Figures:

1. Training/Validation Curves (Include in report)
2. Confusion Matrix (Include in report)
3. Sample Prediction (Include in the report)

Marks: Your report should not be more than 2000 words long. Marks will be allocated in line with the University Common Marking Scheme. Which for the project can be summarized as follows:

Mark Range	Requirements / Achievement
0 – 29 (fail)	Most sections are missing. Dataset is not properly loaded or preprocessed. CNN models are not implemented or not functional. Little evidence of experimentation or evaluation.
30 – 39 (marginal fail)	Most sections attempted but with methodological errors (e.g., incorrect preprocessing, invalid evaluation, or incomplete training). Limited interpretation of results.
40 – 49 (pass)	All required components implemented (data preprocessing, baseline model, enhanced model, evaluation). Methodology is mostly correct, but interpretation of results is minimal or descriptive only.
50 – 59 (2:2)	All sections are completed with correct methodology. Baseline and enhanced CNN models implemented correctly. Training curves, confusion matrix, and evaluation metrics included. Interpretation shows reasonable understanding but lacks depth.
60 – 69 (merit)	All sections are completed with correct methodology. Clear comparison between baseline and enhanced CNN models. Good interpretation of training behavior, confusion matrix, and performance differences. Design decisions are justified.
70 – 100 (distinction)	Excellent implementation and experimentation. Strong justification of architectural and hyperparameter choices. Insightful interpretation of results and limitations. Clear presentation, critical reflection, and evidence of independent thinking and analytical problem-solving skills. References to relevant sources where appropriate.

MOST IMPORTANT OF ALL!!

Read this part very carefully. You must work on this assignment completely on your own. Do not ask classmates for advice and do not share your answers with anybody. If you do look up something online, make sure you include a reference to it. Marks will not be given to work that is clearly not your own.

Statement on the use of AI

For this coursework, the ethical and intentional use of Generative Artificial Intelligence Tools (AI) is permitted (with the exception of the use of AI for the specific purpose of writing or amending code and generate figures, which is not permitted as this assignment tests your ability to write code or visual tools to implement machine learning pipelines).

Whenever AI tools are used you should:

- Cite as a source, any AI tool used in completing your assignment.
- Acknowledge how you have used AI in your work.
- Using AI without citation or against assessment guidelines falls within the definition of plagiarism or cheating, depending on the circumstances, under the current Academic Integrity Policy, and will be treated accordingly. Making false or misleading statements as to the extent, and how AI was used, is also an example of "dishonest practice" under the policy.

For this coursework, the ethical and intentional use of Generative Artificial Intelligence Tools (AI) is permitted at the level of "AI Planning".

AI Planning	AI may be used for pre-task activities such as brainstorming, outlining and initial research. This level focuses on the effective use of AI for planning, synthesis, and ideation, but assessments should emphasize the ability to develop and refine these ideas independently. *You may use AI for planning, idea development, and research. Your final submission should show how you have developed and refined these ideas.*

Please contact Linda, Craig, or Taha for specific guidance.

Submission deadline: May, 8, 2026